

Surrogate-based calibration and prediction of hysteretic model parameters for steel beam-column bolted T-stub connections

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ABSTRACT

The nonlinear behavior of steel beam-column bolted T-stub (BCBT) connections under cyclic loading can potentially be represented using the *Pinching4* model in *OpenSees*. However, its application is limited by the lack of parameter specifications tailored to these connections, particularly for parameters controlling pinching effects and stiffness degradation, which are not directly observable from experimental data. Manual calibration of these parameters is challenging due to the model's implicit nature and high-dimensional parameter space. To address these challenges, this study (1) proposes a novel calibration approach by combining Polynomial Chaos Expansion (PCE) and Genetic Algorithm (GA) to calibrate the *Pinching4* model parameters for tested specimens available in the literature, and (2) develops predictive models for specifying parameters for future use of the *Pinching4* model for steel BCBT connections. Specifically, PCE is utilized to approximate the discrepancy between the *Pinching4* model predictions and experimental data to reduce computational cost, while GA is employed for its robust global search capabilities in optimization tasks. Furthermore, Elastic Net (ENet) regression is used to develop predictive models for the *Pinching4* model parameters for untested connections, effectively handling high-dimensional features and small sample sizes, thereby mitigating multicollinearity and overfitting issues. The resulting regression models demonstrate favorable predictive performance, enhancing the accessibility and practical application of the *Pinching4* model in modeling the moment-rotation hysteretic behavior of untested steel BCBT connections.

1. Introduction

Steel beam-column bolted T-stub (BCBT) connections, as shown in Fig. 1, eliminate the need for welding during installation, facilitating on-site assembly and the replacement of components during maintenance, thus making them highly suitable for pre-fabricated steel structures. Typically classified as partially-restrained (semi-rigid), steel BCBT connections exhibit favorable ductility and energy dissipation capacity [1,2], significantly influencing the overall seismic performance of a structure during earthquakes [3,4]. Hysteretic models, representing their nonlinear moment-rotation behavior under cyclic loading, are essential tools for seismic response analysis and performance-based design of steel building structures containing BCBT connections.

Limited research has focused on developing hysteretic models specifically tailored for steel BCBT connections. Hu et al. [5,6] developed component-based models to approximate the cyclic behavior of BCBT connections by combining the force-deformation behavior of individual components of the connection, such as the T-flange, tension bolts, and T-stem. Nevertheless, these

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mechanics-based engineering models are not able to capture the post-peak strength deterioration and stiffness degradation due to the simplifications or assumptions made for relevant components. In contrast, Leon and Kim [7,8] adopted the pinching model provided in DRAIN-2DX [9] to simulate the cyclic behavior of BCBT connections, in which the cyclic envelopes were determined by curve-fitting to experimental data. However, no detailed hysteretic rules regarding the unloading and reloading paths were provided, limiting its potential applications to other T-stub connections. Recent research shows that the *Pinching4* model in *OpenSees* [10] is effective for accurately simulating the nonlinear behavior of various types of connections under cyclic loading [11–16]. Specifically, the *Pinching4* model can capture pinching effects and account for cyclic degradation in both stiffness and strength, making it a promising tool for modeling the behavior of steel BCBT connections studied in this paper.

However, the current use of the *Pinching4* model lacks parameter specifications tailored for steel BCBT connections. Typically, tests must be conducted first, and the model parameters are then calibrated using experimental data. Due to the highly nonlinear behavior of steel BCBT connections, parameters controlling pinching effect and degradation of stiffness and strength in the *Pinching4* model cannot be directly observed from experiments. Moreover, the versatility of the *Pinching4* model introduces an additional challenge in calibrating its high-dimensional unknown model parameters. Previous attempts to calibrate *Pinching4* model parameters for various structural components under cyclic loading, include empirical approaches for high-strength steel end-plate connections [15] and reinforced concrete columns with plain bars [16]. Generally, these approaches rely on manually extracting parameters (e.g., those controlling cyclic stiffness degradation) from hysteretic loops and then averaging across specimens. Such methods are time-consuming and can introduce subjectivity, limiting their reliability for high-dimensional parameter sets. By contrast, Charters Morais et al. [17] employed a Genetic Algorithm (GA) [18,19] to calibrate the *Pinching4* model for low-rise unreinforced masonry buildings, leveraging GA's robust global search capability and suitability for parallel computing. However, the calibration method in Ref. [17] was tied to specific software (Tremuri) and didn't incorporate hysteretic rule parameters governing unloading and reloading stiffness degradation in the *Pinching4* model, thus restricting broader applications.

To address these gaps, this paper provides a surrogate model-based calibration approach, combining Polynomial Chaos Expansion (PCE) and GA to identify the *Pinching4* model parameters for steel BCBT connections. Among various surrogate modeling techniques, PCE is highly effective for approximating system responses of complex models lacking explicit analytical forms [20,21], significantly reducing computational cost for high-dimensional, multi-query tasks such as optimization [22,23]. GA, inspired by biological evolution, iteratively improves candidate solutions through selection, crossover, and mutation, efficiently searching high-dimensional parameter space [24–26].

Additionally, the absence of predictive equations for *Pinching4* model parameters specific to T-stub connection hinders its practical application when experiments are unavailable. Developing predictive models would enable engineers to estimate them based on T-stub connection's geometric dimensions and material properties. Such predictive models can be constructed by regressing the calibrated *Pinching4* model parameters based on tested steel BCBT connections available in the literature. Due to the complex nature of BCBT connections with multiple interactive components, some features can be unimportant or redundant to be included as predictors in a regression model for certain model parameter, leading to a potential issue of multicollinearity. Furthermore, the limited number of available steel BCBT connection specimens can lead to overfitting of the predictive models. Conventional regression methods, such as Ordinary Least Squares (OLS) regression [27], often struggle with issues of multicollinearity and overfitting. Elastic Net (ENet) regression [28,29], which combines the advantages of Least Absolute Shrinkage and Selection Operator (LASSO) (L1 penalty) [30] as well as Ridge (L2 penalty) [31], offers a promising solution for handling high-dimensional feature sets and small sample sizes. In particular, L1 regularization can shrink coefficients of unimportant or redundant features to zero (feature selection), while L2 regularization reduces the model's sensitivity to multicollinearity and prevents overfitting.

Compared with machine learning methods available for regression tasks, such as tree-based XGBoost [32] and Random Forest [33], and kernel-based Support Vector Regression [34], ENet regression offers a transparent and interpretable “optimal linear model” for each *Pinching4* model parameter, facilitating direct application in structural design and analysis. In contrast, machine learning models often require additional techniques (e.g., SHAP [35]) to enhance interpretability. It should be pointed out that although advanced

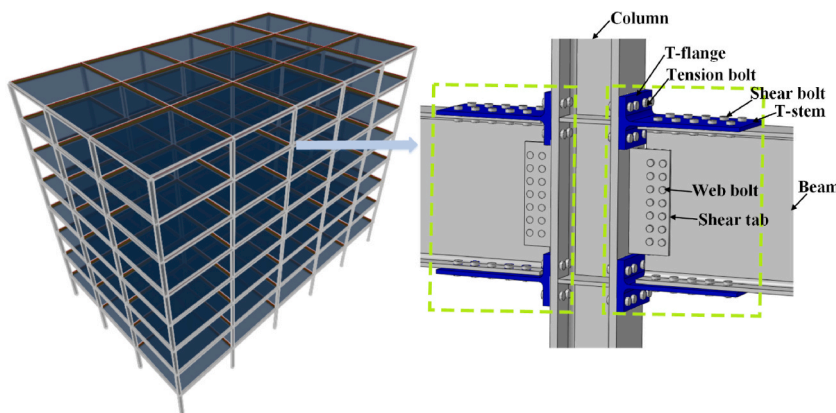


Fig. 1. Typical steel beam-column bolted T-stub (BCBT) connections in a steel frame structure.

machine learning methods have the potential to predict complete hysteretic curves for steel BCBT connections, their high demands for larger training datasets and inherent black-box nature make them less suitable for the scenario investigated in this study.

The remainder of this paper is organized as follows: Section 2 introduces the thirteen *Pinching4* model parameters requiring calibration for BCBT connections. Section 3 explains the calibration approach, combining PCE-based surrogate models and GA optimization for parameter identification. Section 4 develops ENet regression equations to predict the *Pinching4* model parameters for untested steel BCBT connections. Finally, Section 5 concludes this paper.

2. *Pinching4* model parameters considered for calibration

The *Pinching4* model was originally developed to construct a uniaxial material to model a pinched load-deformation relationship for reinforced-concrete beam-column joints under reversed-cyclic loading with degradation in strength and stiffness (both unloading and reloading stiffness) [14]. The *Pinching4* model, implemented in *OpenSees* [10], is used in this study. It consists of a total of 39 parameters (shown in Table A1 in the appendix) and can be reduced to 28 when symmetry is assumed for both positive and negative loading. Of the 28 parameters, 8 parameters pertain to the four critical points on the envelope, 3 parameters are associated with the onset of reloading and strength developed upon unloading to control the pinching shape of the hysteresis loops, and 15 parameters relate to degradation. Specifically, the 15 degradation parameters include 3 groups, 5 for each to consider the cyclic degradations in unloading stiffness, reloading stiffness, and strength, respectively. Note that the damage rule in the *Pinching4* model, which defines cyclic stiffness degradation in reloading, also defines in-cycle strength degradation [36]. In the context of a rotation (displacement)-controlled cyclic loading protocol, in-cycle strength degradation refers to the reduction in strength within the same load step, where subsequent cycles at the same rotation level exhibit decreased strength. Cyclic strength degradation, in comparison, refers to the cumulative damage that occurs across multiple load steps, resulting in a progressive reduction in the peak strength observed in each new loading step [37]. Additionally, the last two parameters relate to damage: one parameter (*dmgType*) indicating the type of damage, either based on energy or load cycles, and the other parameter (*gE*) representing the ratio of maximum energy dissipation under cyclic loading to that under monotonic loading.

As the hysteretic relationship can be defined using either a cyclic envelope or a monotonic backbone [36], a cyclic envelope is adopted for the *Pinching4* model in this study. Consequently, the 5 parameters related to cyclic strength degradation can be disregarded because the cyclic envelope already accounts for these effects. Given that the cyclic envelopes of steel BCBT connections can be readily determined with the predictive four-stage envelope model developed by the authors [38], this envelope model is adopted aiming to focus on the parameters other than the 8 defining the envelope. Further details on the cyclic envelopes are provided in Table A2 of the appendix. In addition, the adopted damage type is based on loading cycles (*dmgType*), and the parameter *gE* is not needed. As a result, out of the 28 parameters, 13 parameters remain to be calibrated.

Among the 13 parameters, the three parameters controlling the pinching shape are critical for steel BCBT connections. Specifically,

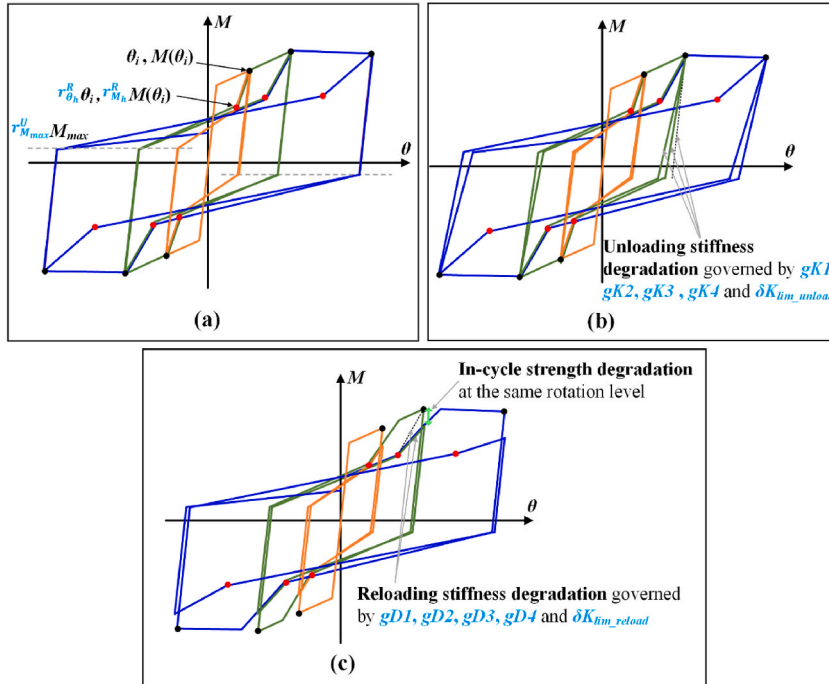


Fig. 2. Illustration of *Pinching 4* hysteresis loops under rotation-controlled cyclic loading with two cycles per load step: (a) no stiffness degradation, (b) with unloading stiffness degradation, and (c) with reloading stiffness degradation (in-cycle strength degradation).

these parameters are: the ratio of the deformation at which reloading begins to the maximum historic deformation ($r_{\theta_h}^R$), the ratio of corresponding force to the force associated with the maximum historic deformation ($r_{M_h}^R$), and the ratio of strength developed upon unloading from negative load to the maximum strength developed under monotonic loading ($r_{M_{max}}^U$). These parameters are illustrated in Fig. 2(a), where θ_i denotes the rotation defining the end of the reloading cycle for each load step, namely the unloading point (indicated by the black dots) in rotation-controlled cyclic loading. $M(\theta_i)$ represents the moment corresponding to θ_i , and M_{max} is the maximum moment on the cyclic envelope. Note that the cyclic behavior of steel BCBT connections is considered symmetric in both positive and negative directions [34].

Additionally, five parameters govern cyclic stiffness degradation during unloading: $gK1$, $gK2$, $gK3$, $gK4$, and δK_{lim_unload} . Another set of five parameters control cyclic stiffness degradation during reloading (in-cycle strength degradation): $gD1$, $gD2$, $gD3$, $gD4$, and δK_{lim_reload} . Specifically, $gK1$ through $gK4$ are coefficients in the function that defines the unloading stiffness damage index δK_{i_unload} , while $gD1$ through $gD4$ are coefficients in the function that defines the reloading stiffness damage index δK_{i_reload} . The limit of unloading stiffness degradation, δK_{lim_unload} , should satisfy constraints associated with $gK1$ to $gK4$, ensuring that $\delta K_{i_unload} = f(gK1, gK2, gK3, gK4) \leq \delta K_{lim_unload}$ [10]. Similarly, the limit of reloading stiffness degradation, δK_{lim_reload} , should satisfy constraints associated with $gD1$ to $gD4$, ensuring that $\delta K_{i_reload} = f(gD1, gD2, gD3, gD4) \leq \delta K_{lim_reload}$ [10]. Fig. 2(b) and (c) illustrate the cyclic degradation in unloading stiffness, cyclic degradation in reloading stiffness (in-cycle strength degradation), respectively.

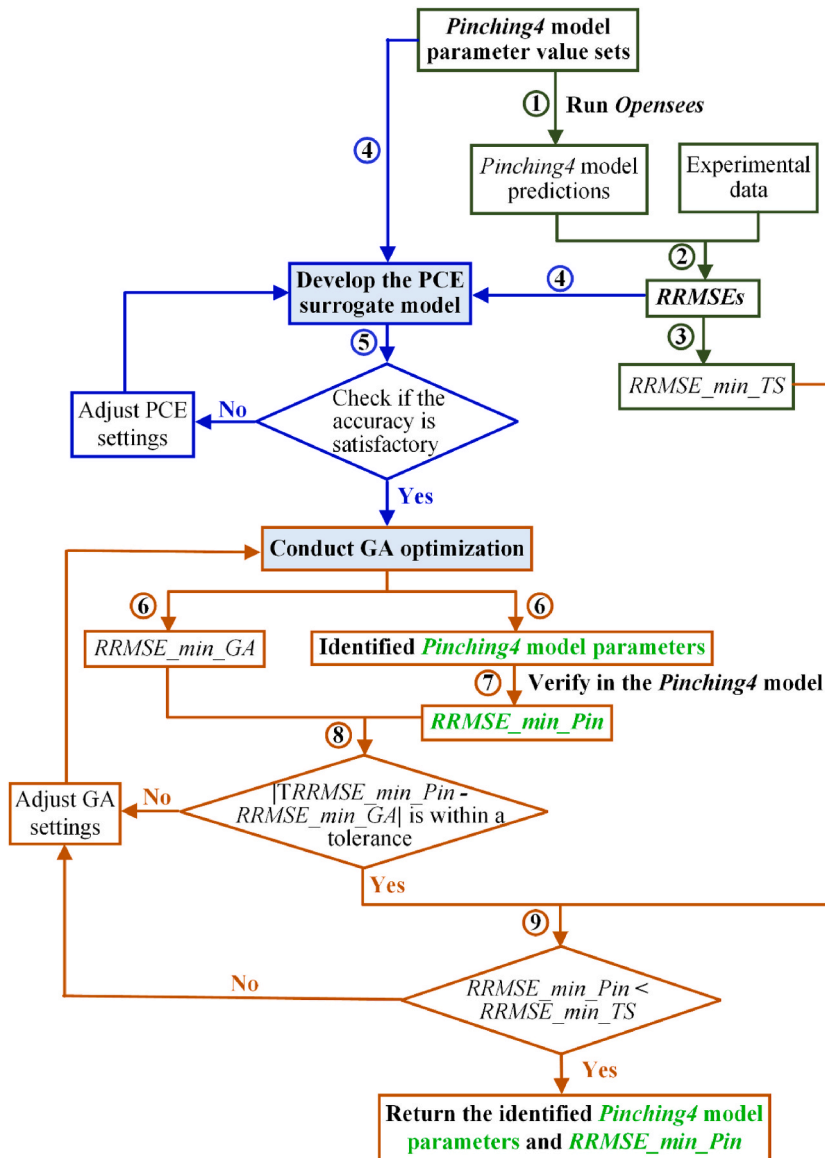


Fig. 3. The procedure of parameter calibration of the *Pinching4* model for each specimen.

3. Calibration of *Pinching4* model parameters via PCE and GA

3.1. Procedure for parameter calibration

The 13 parameters of the *Pinching4* model are calibrated for each of the 20 steel BCBT connections subjected to cyclic loading. The details of these specimens have been previously reported in Ref. [38]. Given the implicit nature of the *Pinching4* model and its high-dimensional parameter space, a surrogate model-based optimization approach is adopted. This approach integrates PCE for approximating the *Pinching4* model's response and GA for performing global optimization. The purpose of the calibration is to identify the optimal set of parameter values that minimize the discrepancy (measured by the relative root mean square error, *RRMSE*) between *Pinching4* model predictions and experimental hysteresis. The procedure for calibrating the *Pinching4* model parameters for each specimen is illustrated in Fig. 3, which involves three main steps.

The first step is to generate samples for training the PCE surrogate models. Among the 13 model parameters to be calibrated, 11 are considered mutually independent, while the remaining two parameters, i.e., δK_{lim_unload} and δK_{lim_reload} , are subject to inequality constraints depending on the other parameters. To ensure that all training samples satisfy these constraints, conditional sampling is employed. Specifically, Latin Hypercube Sampling (LHS) [39] is utilized to generate 20000 samples for the 11 independent parameters, each drawn from their respective prior distributions (as presented in Table A3 in the appendix). LHS ensures uniform coverage across the full distribution range of each parameter, enhancing sampling efficiency. Using these samples, the lower bounds δK_{i_unload} and δK_{i_reload} are calculated. Subsequently, δK_{lim_unload} and δK_{lim_reload} are conditionally sampled by defining their prior distributions truncated at these lower bounds. For each sample, the truncation points for δK_{lim_unload} and δK_{lim_reload} are set to their respective lower bounds, and samples are drawn from the resulting truncated prior distributions. Adhering to these constraints is essential to ensure the reliability of the calibrated model parameters without compromising the accuracy of the calibration. Conversely, neglecting these constraints could lead to physically unreasonable predictions of hysteretic behavior (e.g., unloading stiffness might increase with cumulative cyclic damage), resulting in significant deviations from the experimental curves.

The sample size for training the PCE surrogate model is determined by ensuring that it is no less than the total number of polynomial terms in the PCE model [20]. A TCL script is used to call *OpenSees* to generate 20000 hysteretic curves predicted by the *Pinching4* model for each specimen. These predictions are then compared with the experimental hysteretic curve (no need for amplitude normalization) of the specimen to calculate 20000 *RRMSE* values. The preliminary minimum *RRMSE* among the training samples (TS), denoted as *RRMSE_min_TS*, can be directly extracted in this process, without requiring parameter identification through other methods.

The second step is to develop a PCE Surrogate model for each specimen using the training samples generated in previous step. In the PCE surrogate model, the *Pinching4* model parameters serve as the input variables, while the calculated *RRMSE* are considered the response variable. The error between the predictions of the PCE surrogate model and the *RRMSE*s of the training samples is then evaluated. If the accuracy of the surrogate model is unsatisfactory, the PCE settings (e.g., polynomial degree and truncation options) are adjusted, and the model is reconstructed.

The third step is to conduct GA optimization based on the constructed PCE surrogate mode. GA is utilized to find the minimum *RRMSE* (denoted by *RRMSE_min_GA*) between the PCE surrogate model predictions and the experimental results, along with the corresponding *Pinching4* model parameters. The *Pinching4* model parameters identified by GA are subsequently evaluated by comparing the *Pinching4* model predictions with experimental data, obtaining the *RRMSE* (denoted by *RRMSE_min_Pin*). The discrepancy between *RRMSE_min_Pin* and *RRMSE_min_GA* should not exceed a tolerance level, such as 5 % as considered in this study. Additionally, the *RRMSE_min_Pin* should be less than *RRMSE_min_TS*.

Details on the development of PCE surrogate models and implementation of GA optimization are provided in the following two sections.

3.2. PCE surrogate models

Define a random vector $\mathbf{x}$ to represent the *Pinching4* model parameters, $\mathbf{x} = (x_1, \dots, x_{13})$, where $x_1, \dots, x_{13}$ corresponds to $r_\theta, \dots, \delta K_{lim_reload}$, respectively. To use Hermite polynomials [40] as the basis functions of the PCE, the distribution of each component x_i ($i = 1, \dots, 13$) of the vector, as shown in Table A3 in the appendix, needs to be transformed into a standard normal distribution. The transformed vector is denoted as $\mathbf{z}$, with $\mathbf{z} = (z_1, \dots, z_{13})$. *RRMSE* predicted by the PCE surrogate model can be expressed as:

$$\begin{aligned} RRMSE_{PCE}(\mathbf{z}) = & c_0 H_0 + \sum_{i=1}^n \left(\sum_{\alpha_i=1}^{\infty} c_{\alpha_i} H_{\alpha_i}(z_i) \right) + \sum_{i=1}^{n-1} \sum_{j=i+1}^n \left(\sum_{\alpha_i=1}^{\infty} \sum_{\alpha_j=1}^{\infty} c_{\alpha_i, \alpha_j} H_{\alpha_i}(z_i) H_{\alpha_j}(z_j) \right) + \dots \\ & + \sum_{\alpha_1=1}^{\infty} \sum_{\alpha_2=1}^{\infty} \dots \sum_{\alpha_n=1}^{\infty} c_{\alpha_1, \alpha_2, \dots, \alpha_n} H_{\alpha_1}(z_1) H_{\alpha_2}(z_2) \dots H_{\alpha_n}(z_n) \end{aligned} \quad (1)$$

where c_0 is the constant term; c_{α_i} , c_{α_i, α_j} , ..., and $c_{\alpha_1, \alpha_2, \dots, \alpha_n}$ are the expansion coefficients; $n = 13$ is the number of model parameters; $H_{\alpha_i}(z_i)$, $H_{\alpha_j}(z_j)$, ..., and $H_{\alpha_n}(z_n)$, are univariate Hermite polynomials (basis functions), with the subscripts α_i , α_j , ..., and α_n representing the polynomial indices; and $H_0 = 1$ is the zero-indices Hermite polynomial.

Although Hermite polynomials can theoretically extend to positive infinity and Eq. (1) contains an infinite number of terms, high-order terms are usually truncated in practice. Assuming τ is the highest order retained after truncation, the total number of PCE terms

($T+1$) after truncation is:

$$T+1 = \frac{(\tau+n)!}{(\tau!n!)} \quad (2)$$

where T is the total number of polynomial terms excluding the constant term. In this study, τ is set from 1 to 5 and the maximum total number of polynomial terms in the PCE model is $(5+13)!/(5!13!) = 8568$. The training sample size of 20000 is approximately 2.3 times the maximum total number of polynomial terms.

3.2.1. Calculation of the PCE coefficients

Given the large sample size of 20000 used to train the PCE surrogate model, the OLS method, which is suitable for large datasets, is employed to estimate the PCE coefficients. For each training sample ($i = 1, \dots, N$, with $N = 20000$), the transformed input vector $\mathbf{z}^{(i)} = (z_1^{(i)}, \dots, z_3^{(i)})$ are input variables, and the $RRMSE$ between the *Pinching4* model prediction and experimental result is “observed” response variable, denoted as $RRMSE_{obs}(\mathbf{z})$. Thus, the training dataset is given as $\{\mathbf{z}^{(i)}, RRMSE_{obs}(\mathbf{z}^{(i)})\}$. To estimate the PCE coefficients, a design matrix $\mathbf{A}$ of dimensions $N \times (T+1)$ is constructed. The design matrix is based on the values of the basis functions evaluated at each training sample. The element A_{ij} in the i -th row and j -th column of the matrix $\mathbf{A}$ is defined as follows.

● **For the constant term ($j = 1$)**

$$A_{ij} = H_0(\mathbf{z}^{(i)}) = 1 \quad (3)$$

● **For the non-constant terms ($j = 2, 3, \dots, T+1$)**

$$A_{ij} = H_{\alpha_j}(\mathbf{z}^{(i)}) = \prod_{k=1}^{13} H_{\alpha_{kj}}(z_k^{(i)}) \quad (4)$$

where $\alpha_j = (\alpha_{1,j}, \alpha_{2,j}, \dots, \alpha_{13,j})$ are the multi-indices corresponding to the orders of the Hermite polynomials for each input variable in the j -th term.

The arrangement of the α_j is systematic.

(1) **Univariate terms:** All combinations where only one α_k is non-zero (from 1 up to τ) and the rest are zero.

This generates the first set of columns after the constant term. For example, columns 2 through 14 of matrix $\mathbf{A}$ correspond to univariate terms, with indices $(\alpha_{1,2} = 1, \alpha_{2,2} = 0, \dots, \alpha_{13,2} = 0)$, $(\alpha_{1,3} = 0, \alpha_{2,3} = 1, \dots, \alpha_{13,3} = 0)$, $\dots$, $(\alpha_{1,14} = 0, \alpha_{2,14} = 0, \dots, \alpha_{13,14} = 1)$ in sequence.

(2) **Multivariate terms:** All possible combinations where the sum of the α_k indices is between 2 and τ , ensuring that the total polynomial order does not exceed τ .

Following this arrangement, all Hermite polynomial terms up to the specified maximum order τ are included, capturing the necessary interactions between input variables. The complete design matrix $\mathbf{A}$ can thus be expressed as:

$$\mathbf{A} = \begin{bmatrix} H_{\alpha_1}(\mathbf{z}^{(1)}) & H_{\alpha_2}(\mathbf{z}^{(1)}) & H_{\alpha_3}(\mathbf{z}^{(1)}) & \dots & H_{\alpha_{T+1}}(\mathbf{z}^{(1)}) \\ H_{\alpha_1}(\mathbf{z}^{(2)}) & H_{\alpha_2}(\mathbf{z}^{(2)}) & H_{\alpha_3}(\mathbf{z}^{(2)}) & \dots & H_{\alpha_{T+1}}(\mathbf{z}^{(2)}) \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ H_{\alpha_1}(\mathbf{z}^{(N)}) & H_{\alpha_2}(\mathbf{z}^{(N)}) & H_{\alpha_3}(\mathbf{z}^{(N)}) & \dots & H_{\alpha_{T+1}}(\mathbf{z}^{(N)}) \end{bmatrix} \quad (5)$$

Next, the “observed” response values are then organized into the target vector $\mathbf{Y}$:

$$\mathbf{Y} = \begin{bmatrix} RRMSE_{obs}(\mathbf{z}^{(1)}) \\ RRMSE_{obs}(\mathbf{z}^{(2)}) \\ \vdots \\ RRMSE_{obs}(\mathbf{z}^{(N)}) \end{bmatrix} \quad (6)$$

The objective is to determine the PCE coefficient vector $\mathbf{c}_\alpha$ that minimizes the difference between the observed responses and the PCE predictions:

$$\mathbf{c}_\alpha^* = \underset{\mathbf{c}_\alpha}{\operatorname{argmin}} \sum_{i=1}^N (RRMSE_{PCE}(\mathbf{z}^{(i)}) - RRMSE_{obs}(\mathbf{z}^{(i)}))^2 \quad (7)$$

Since $RRMSE_{PCE}(\mathbf{z}^{(i)}) = \mathbf{A}_{i,:} \mathbf{c}_\alpha$, where $\mathbf{A}_{i,:}$ is the i -th row of $\mathbf{A}$, the problem can be reformulated in matrix notation as:

$$\mathbf{c}_\alpha^* = \underset{\mathbf{c}_\alpha}{\operatorname{argmin}} \|\mathbf{Y} - \mathbf{A}\mathbf{c}_\alpha\|_2^2 \quad (8)$$

The standard solution to this least squares problem is:

$$\mathbf{c}_\alpha^* = (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T \mathbf{Y} \quad (9)$$

where $\mathbf{A}^T$ is the transpose of the design matrix $\mathbf{A}$, and $(\mathbf{A}^T \mathbf{A})^{-1}$ is the inverse of $\mathbf{A}^T \mathbf{A}$.

After obtaining the PCE coefficients $\mathbf{c}_\alpha^*$, the PCE model can be used to make predictions in the transformed input space. However, to interpret these results in terms of the original input vector $\mathbf{x}$, an inverse transformation should be applied. This involves mapping the standardized vector $\mathbf{z}$ back to the original vector $\mathbf{x}$, allowing to compute the PCE predictions $RRMSE_{PCE}(\mathbf{x}^{(i)})$ for the original input data.

3.2.2. Assessment of the PCE surrogate models

The accuracy of the developed PCE surrogate models is assessed using both leave-one-out cross-validation error ($LOOCV_err$) [41] and residual (ϵ_i) plots. The $LOOCV_err$ is a comprehensive measure of predictive error, achieved by sequentially leaving out one data point, constructing the model with the remaining data and predicting the value of the omitted point. Fig. 4 displays the $LOOCV_err$ curves with respect to different τ (the highest order of Hermite polynomials). It is observed that the optimal truncated order for different specimens, which balances the complexity and predictive accuracy, is either 3 or 4. Beyond this order, further improvement in $LOOCV_err$ is minimal, and the error even tends to increase due to overfitting. It indicates that the selected range of polynomial orders in constructing the PCE models is appropriate. Additionally, $LOOCV_err$ varies among different specimens, which may be attributed to the inherent data characteristics of each specimen. Table 1 presents the $LOOCV_err$ values corresponding to the optimal τ for different PCE surrogate models. These values range from 0.035 % to 0.064 % across all specimens, with an average error of 0.047 %. No extreme error values are observed. These results demonstrate that the PCE models exhibit stable predictive performance across different specimens and achieve high accuracy.

Residual plots visually depict the differences between the predicted and “actual” values. Fig. 5(a) and (b) show the residual plots for the PCE surrogate models with the lowest and highest $LOOCV_err$ values, respectively. The residual plots for other PCE models fall between these two extremes. As seen in the plots, the residuals are symmetrically distributed around $\epsilon_i = 0$, with no apparent bias toward either side and no significant outliers. The residuals’ fluctuations remain constant across the entire data range, exhibiting homoscedasticity. These results further confirm the accuracy of the PCE surrogate models.

3.3. GA optimization and model calibration results

In GA optimization, the optimization goal is to identify the set of *Pinching4* model parameter values that minimize $RRMSE_{PCE}(\mathbf{z})$ predicted by the PCE surrogate model. The fundamental implementation process of GA optimization is illustrated in Fig. 6. The performance of GA is highly sensitive to several key parameters: population size, crossover probability, mutation probability, and elite preservation strategy [18,42]. Specifically: population size sets the number of individuals in each generation; crossover probability indicates the likelihood that individuals will undergo crossover to produce offspring; mutation probability defines the chance that individuals will undergo mutation; and elite preservation strategy involves retaining a number of top-performing individuals unchanged into the next generation. To ensure optimal performance in this study, a grid search approach is employed to systematically determine the optimal combination of these parameters. The ranges and intervals used in the grid search are specified in Table 2.

The maximum number of generations is set to 200, defining the maximum iterations GA will run before terminating. To enhance computational efficiency, GA optimization leverages parallel computing, significantly reducing computation time by distributing the workload across multiple processors. Additionally, random uniform selection, scattered crossover, and Gaussian mutation are

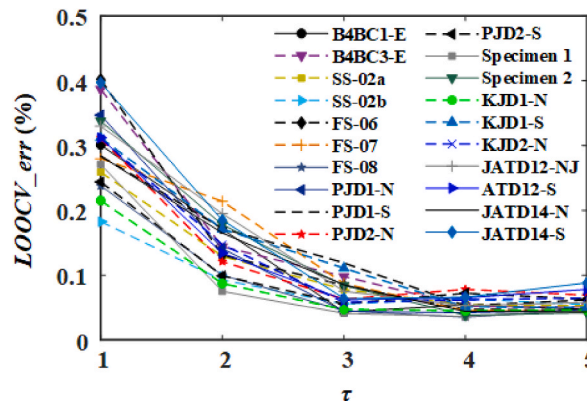


Fig. 4. $LOOCV_err$ curves for different τ

Table 1

$LOOCV_{err}$ values corresponding to the optimal τ for different PCE surrogate models.

Specimen	$LOOCV_{err}$ (%)	Specimen	$LOOCV_{err}$ (%)
B4BC1-E	0.043	PJD2-S	0.056
B4BC3-E	0.052	Specimen 1	0.036
SS-02a	0.054	Specimen 2	0.038
SS-02b	0.056	KJD1-N	0.047
FS-06	0.047	KJD1-S	0.051
FS-07	0.045	KJD2-N	0.058
FS-08	0.035	JATD12-N	0.053
PJD1-N	0.041	JATD12-S	0.062
PJD1-S	0.053	JATD14-N	0.044
PJD2-N	0.063	JATD14-S	0.064

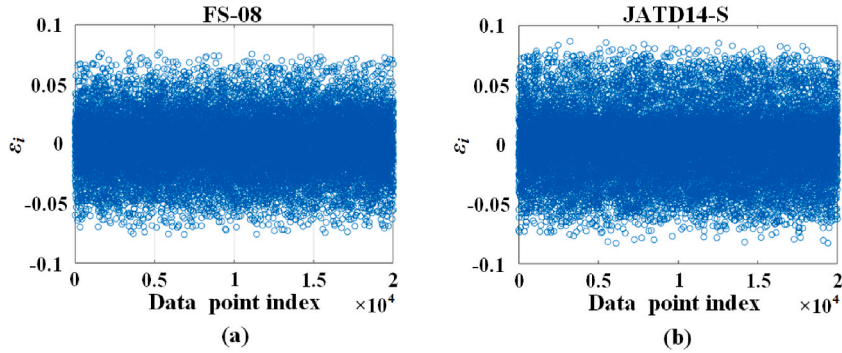


Fig. 5. Residual plots for PCE surrogate models with the lowest (a) and highest (b) $LOOCV_{err}$ values.

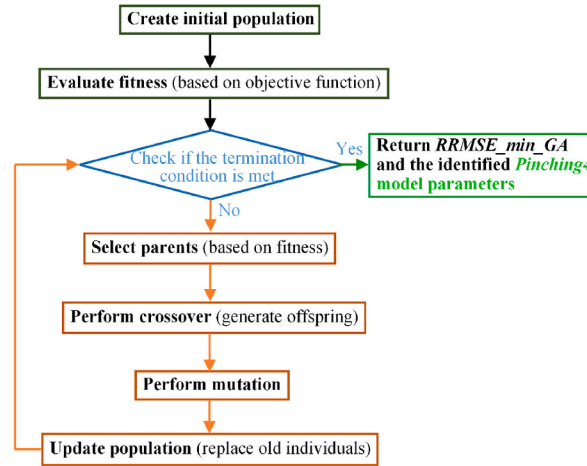


Fig. 6. The process of GA optimization.

Table 2

GA parameter ranges for grid search.

Parameter	Values or Range	Interval
Population size	$50 \times 2^{(n-1)}$, where $n = 1, 2, 3, 4$	–
Crossover probability	0.6–0.9	0.1
Mutation probability	0.01–0.1	0.02
Elite count	1 %–10 % of population size	2 %

employed in the GA. Random uniform selection provides equal selection probability to all individuals, maintaining population diversity. Scattered crossover generates offspring by combining genes from parents in varied patterns, promoting genetic diversity. Gaussian mutation introduces small, normally distributed changes to individuals, helping to avoid local optima.

For each specimen, GA was executed multiple times independently to ensure the repeatability and consistency of the optimization results. The best objective function value ($RRMSE_min_GA$) obtained through GA was then compared with the smallest $RRMSE$ ($RRMSE_min_TS$) derived from the training samples used for constructing the PCE surrogate model. As shown in Fig. 7, $RRMSE_min_GA$ consistently outperforms $RRMSE_min_TS$ across all specimens, further validating the effectiveness and robustness of GA optimization.

After GA searches for $RRMSE_min_GA$ using the PCE surrogate model, it simultaneously outputs the corresponding *Pinching4* model parameters, which are taken as the calibrated results. The detailed values of the *Pinching4* model parameters, along with the resulting $RRMSE_min_Pin$ (calculated in the *Pinching4* model), are presented in Table 3. It is observed that all calibrated results fall within the theoretical bounds of the *Pinching4* model parameters. For example, $r_{\theta_h}^R$, $r_{M_h}^R$, and $r_{M_{max}}^U$ range from 0.16 to 0.71, 0.32 to 0.77, and 0.09 to 0.47, respectively. These three parameters have explicit physical interpretations: $r_{\theta_h}^R$ accounts for cyclic friction effects and the release of pretension force resulting from material yielding; $r_{M_h}^R$ captures slipping between the T-stem and beam flange; and $r_{M_{max}}^U$ accounts for the influence of reloading stiffness reduction resulting from slipping as well as bolt movement on the T-flange. Similarly, the two parameters δK_{lim_unload} and δK_{lim_reload} , which control the limits of unloading and reloading stiffness degradation, range from 0.10 to 0.34 and 0.09 to 0.33, respectively, and the eight parameters ($gK1$, ..., $gD4$) defining stiffness damage indices vary between -1.17 and 2.52. In addition, the significant variations in the *Pinching4* model parameters across different specimens indicate that these parameters are highly dependent on the physical characteristics (geometric dimensions and material properties) of the steel BCBT connections.

For 15 out of 20 specimens (excluding KJD1-N, KJD1-S, JATD12-S, JATD14-N, and JATD14-S), the $RRMSE_min_Pin$ values range from 0.091 to 0.195, indicating an adequate fit of the *Pinching4* model predictions to the experimental hysteretic curves. This is further evidenced in Fig. 8, which compares the hysteretic curves generated by the calibrated *Pinching4* model (black solid lines) with the experimental hysteretic curves (blue solid lines), with the cyclic envelope represented by the yellow dashed lines. It should be pointed out that the *Pinching4* model uses loading protocols extracted from the experimental hysteretic curves. This approach eliminates errors introduced by the loading protocols, focusing instead on the *Pinching4* model's ability to simulate hysteretic behavior.

It is observed that the *Pinching4* model accurately captures key cyclic responses, such as pinching effects, unloading stiffness degradation, and reloading stiffness degradation for the 15 specimens. For the remaining five specimens, the $RRMSE_min_Pin$ ranges from 0.233 to 0.289. The reduction in prediction accuracy is attributed to relatively large differences between the positive and negative directions of the experimental hysteretic curves, as illustrated in Fig. 8. These discrepancies may result from either unexpected test conditions or significant noise present in the experimental data.

4. Predictive modeling of *Pinching4* model parameters using ENet regression

4.1. Methodology of ENet regression for parameter estimation

ENet regression is a computationally efficient method that meets the demands of accuracy, interpretability, and engineering applicability for the data-limited scenario investigated in this study. In ENet regression, the features of steel BCBT connections (geometric dimensions and material properties) serve as input variables, while the calibrated *Pinching4* model parameters are the output variables. The ENet regression formula for a *Pinching4* model parameter (x_i) is expressed as:

$$x_{i,pre}^{(j)} = \beta_{0+} + \sum_{k=1}^p \beta_k u_k^{(j)} \quad (10)$$

where $x_{i,pre}^{(j)}$ denotes the predicted value of the *Pinching4* model parameter x_i for the j -th regression sample (specimen) using ENet

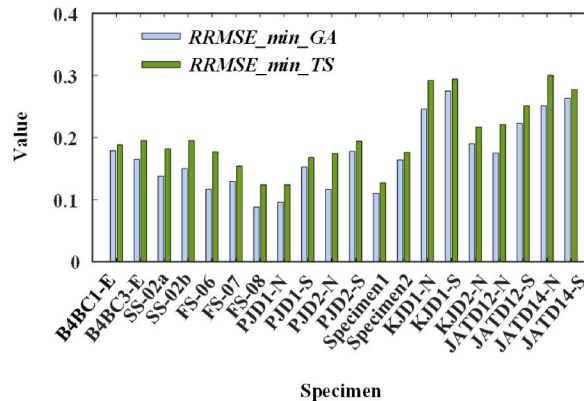


Fig. 7. Comparison of $RRMSE_min_GA$ against $RRMSE_min_TS$.

Table 3

Pinching4 model parameters along with the resulting *RRMSE_min_Pin* for different specimens.

Specimen	$r_{\theta_h}^R$	$r_{M_h}^R$	$r_{M_{max}}^U$	<i>gK1</i>	<i>gK2</i>	<i>gK3</i>	<i>gK4</i>	$\delta K_{lim-unload}$	<i>gD1</i>	<i>gD2</i>	<i>gD3</i>	<i>gD4</i>	$\delta K_{lim-reload}$	<i>RRMSE_min_Pin</i>
B4BC1-E	0.71	0.32	0.09	1.60	2.52	1.63	−1.23	0.30	2.09	0.75	1.34	0.15	0.23	0.184
B4BC3-E	0.65	0.46	0.13	0.94	0.25	1.04	0.55	0.34	0.82	1.28	1.28	0.53	0.21	0.161
SS-02a	0.70	0.67	0.24	0.62	0.39	0.44	0.31	0.19	−0.43	−0.71	1.36	1.94	0.26	0.145
SS-02b	0.61	0.59	0.23	0.35	0.60	0.72	0.20	0.21	−0.35	−0.63	0.93	1.53	0.25	0.153
FS-06	0.43	0.69	0.28	1.09	0.95	−0.23	1.96	0.16	1.26	0.79	1.35	0.98	0.15	0.12
FS-07	0.58	0.71	0.30	0.89	0.82	0.27	1.83	0.18	0.78	1.02	1.12	0.75	0.20	0.132
FS-08	0.16	0.77	0.47	1.13	1.03	−0.17	0.73	0.16	1.14	0.55	0.83	1.01	0.17	0.091
PJD1-N	0.31	0.77	0.10	0.64	0.52	0.42	0.77	0.25	0.97	1.15	0.22	0.82	0.15	0.099
PJD1-S	0.33	0.73	0.09	0.81	0.70	0.31	0.86	0.23	0.50	0.62	0.36	0.72	0.14	0.157
PJD2-N	0.24	0.66	0.12	0.54	0.79	0.66	0.55	0.22	0.55	0.42	0.44	0.69	0.13	0.121
PJD2-S	0.39	0.62	0.11	0.66	0.61	0.53	0.53	0.24	0.85	0.56	0.11	0.73	0.16	0.186
Specimen1	0.67	0.69	0.09	0.41	1.02	0.73	1.09	0.10	1.04	−0.41	−0.74	0.59	0.11	0.113
Specimen2	0.59	0.72	0.10	0.34	1.60	0.78	0.44	0.15	0.91	−0.23	−1.17	−0.44	0.09	0.162
KJD1-N	0.41	0.59	0.11	0.49	0.68	0.42	0.58	0.19	0.62	0.62	0.11	0.83	0.27	0.256
KJD1-S	0.38	0.52	0.14	0.74	0.51	0.49	0.65	0.24	0.37	0.99	0.63	1.27	0.30	0.289
KJD2-N	0.48	0.69	0.10	0.82	0.88	0.81	0.42	0.32	0.75	0.74	0.74	0.88	0.13	0.195
JATD12-N	0.24	0.66	0.12	0.59	1.20	0.40	1.01	0.19	0.52	0.62	0.65	1.10	0.28	0.182
JATD12-S	0.25	0.64	0.09	0.55	0.97	0.49	0.73	0.22	0.58	0.68	0.31	0.95	0.33	0.233
JATD14-N	0.21	0.71	0.14	0.76	0.90	0.94	0.95	0.24	0.81	0.40	0.22	0.63	0.23	0.261
JATD14-S	0.19	0.68	0.13	0.56	0.99	0.33	0.45	0.21	0.92	0.88	0.81	0.77	0.19	0.272

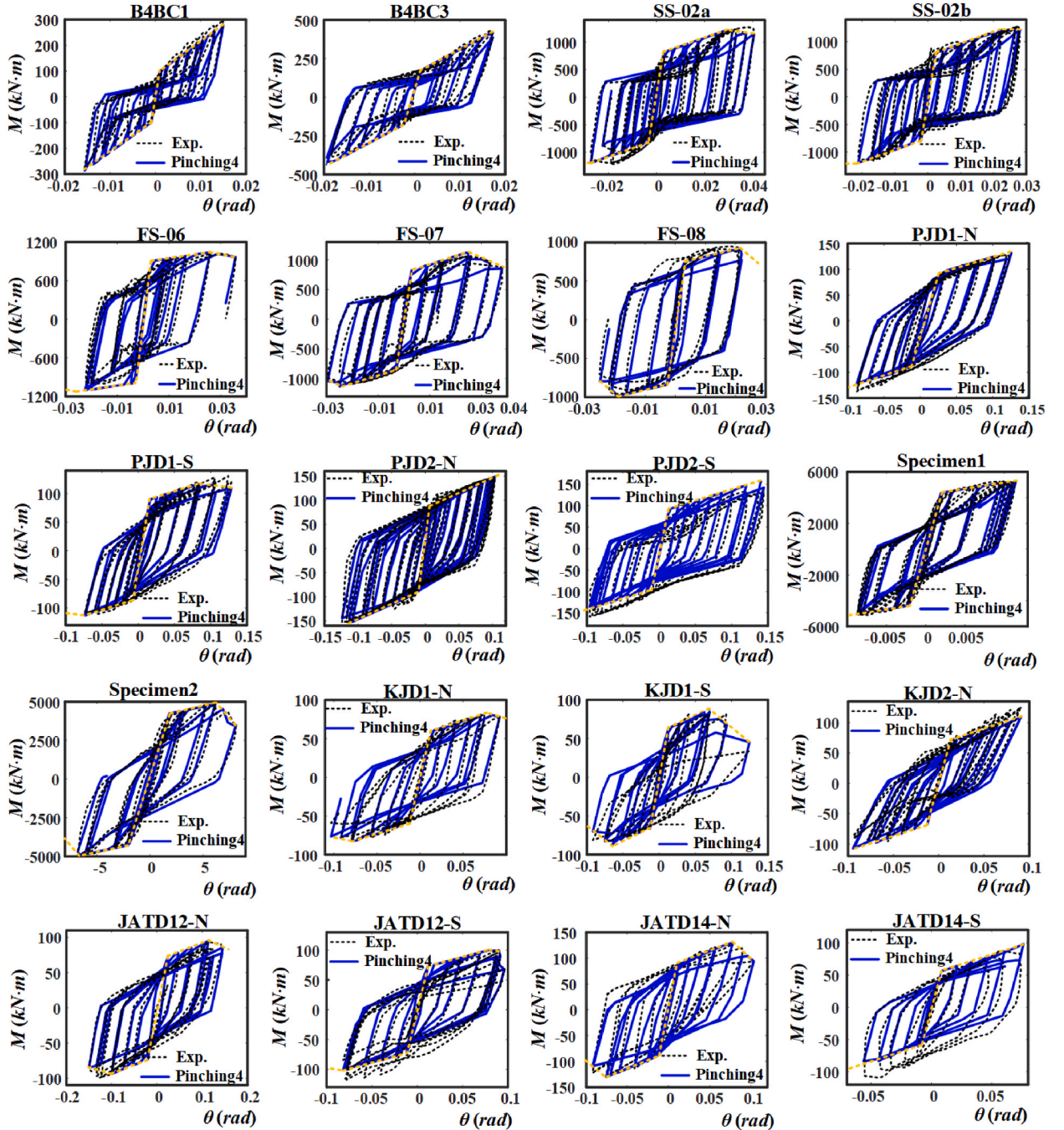


Fig. 8. Comparison between the hysteretic curves generated by the calibrated Pinching4 model and experimental results.

regression; β_0 is the constant term in the regression formula; β_k represents the regression coefficient of the k -th feature; p is the number of features; and $u_k^{(j)}$ is the value of the k -th feature for the j -th regression sample.

The ENet regression loss function combines the OLS residual sum of squares (RSS) with both L1 regularization and L2 regularization terms, expressed as:

$$L_{ENet} = RSS + \lambda \varphi \|\beta\|_1 + \lambda(1 - \varphi) \|\beta\|_2^2 \quad (11)$$

$$RSS = \sum_{j=1}^{n_r} \left(x_{i,obs}^{(j)} - x_{i,pre}^{(j)} \right)^2 \quad (12)$$

$$\|\beta\|_1 = \sum_{k=1}^p |\beta_k| \quad (13)$$

$$\|\beta\|_2^2 = \sum_{k=1}^p \beta_k^2 \quad (14)$$

where n_r is the number of regression samples for a given *Pinching4* model parameter; $x_{i,obs}^{(j)}$ is the calibrated model parameter; β is the vector of regression coefficients; $\|\beta\|_1$ denotes the L1 norm (sum of the absolute values of the regression coefficients); $\|\beta\|_2^2$ denotes the squared L2 norm (sum of the squares of the regression coefficients). The parameter φ balances the L1 and L2 regularization terms, while λ controls the overall regularization strength.

The optimal ENet regression coefficients (β_{ENet}^*) can be determined by minimizing the loss function:

$$\beta_{ENet}^* = \underset{\beta}{\operatorname{argmin}} (RSS + \lambda \varphi \|\beta\|_1 + \lambda (1 - \varphi) \|\beta\|_2^2) \quad (15)$$

Using Lagrange multipliers, it can be shown that the optimization problem in Eq. (15) is equivalent to:

$$\underset{\beta}{\min} (RSS) \text{ subject to } \varphi \|\beta\|_1 + (1 - \varphi) \|\beta\|_2^2 \leq C \quad (16)$$

where C is a constant determined by the constraint. By adjusting λ , a corresponding C can be found such that when $\varphi \|\beta\|_1 + (1 - \varphi) \|\beta\|_2^2$ equals C , the solutions to Eqs. (15) and (16) are identical. This formulation converts the penalized loss function into a constrained optimization problem.

Fig. 9 illustrates the constraint region of the ENet regression and the optimal solutions for a two-dimensional example. In the figure, the contour lines of the RSS form ellipses centered around the optimal solution of the OLS regression (β_{OLS}^*). The points where the RSS contour lines intersect with the constraint region represent the optimal solution for the ENet regression coefficient vector.

When $\varphi = 1$, ENet regression reduces to LASSO regression, which imposes an L1-norm constraint (diamond shape). The optimal solution (β_{LASSO}^*) often lies on an axis, forcing some coefficients to be zero, thus enabling feature selection. When $\varphi = 0$, ENet regression simplifies to Ridge regression, which uses an L2-norm constraint (circular shape). The solution (β_{Ridge}^*) typically does not fall on an axis, so coefficients are shrunk but not driven to zero, reducing multicollinearity and overfitting. When $0 < \varphi < 1$, ENet regression combines L1 and L2 penalties (a blend of diamond and circle). Some coefficients become zero (feature selection), while others are shrunk to handle multicollinearity.

Notably, separate regression models are established for each *Pinching4* model parameters. This allows ENet regression to handle multicollinearity and perform variable selection individually for each parameter. Since both $\delta K_{lim,unload}$ and $\delta K_{lim,reload}$ must satisfy inequality constraints, and ElasticNet regression does not inherently support constraints on the output variables, the regression coefficients for these parameters are manually fine-tuned to ensure compliance.

4.2. Features and regularization parameters for ENet regression

Based on the observed failure modes of the compiled steel BCBT connections [38], and considering the most relevant features for predicting the maximum cyclic moment resistance [38,43–45] and initial rotational stiffness [38,46–49], a total of 9 features were selected. These features are: number of tension bolts on a T-stub (n_t), tension bolt shank diameter (d_{tb}), ultimate tensile strength of the bolt (f_{tbo}), T-flange thickness (t_{TF}), yield strength of the T-stub (f_{yT}), the distance from the T-stem face to the centerline of the tensile bolt hole in the first row (m_0), the distance (in the direction perpendicular to shear force) between shear bolt rows (g_s), beam depth (D_b), and yield strength of the beam (f_{yb}). The statistical values of these features, including maximum (*Max*), minimum (*Min*), mean (μ) and standard deviation (σ), are presented in Table 4.

For the two regularization parameters (φ and λ), φ is uniformly divided into 51 values between 0 and 1 (at intervals of 0.02). The value of λ is determined according to Ref. [50] using the following method:

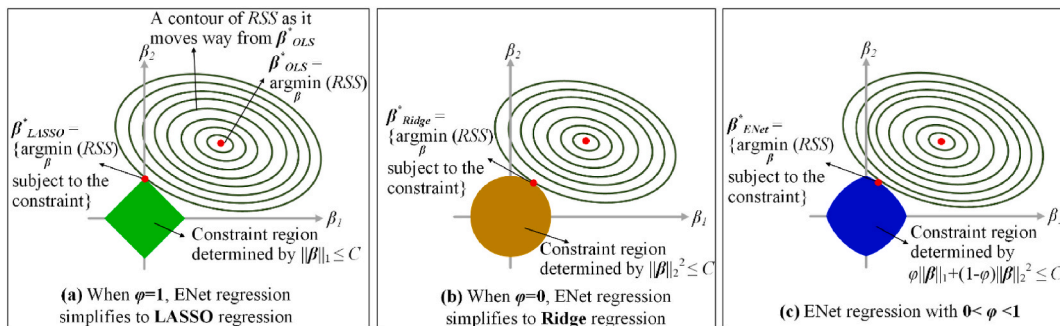


Fig. 9. Constraint regions of ENet regression and optimal solutions for a two-dimensional example.

Table 4

Summary of the feature statistics.

Statistic	n_t	d_{th} (mm)	f_{ubo} (MPa)	t_{Tf} (mm)	f_{YT} (MPa)	m_0 (mm)	g_s (mm)	D_b (mm)	f_{yb} (mm)
Max	8.0	35.0	1246.0	50.0	441.0	88.0	165.1	911.0	423.0
Min	4.0	19.1	606.7	12.0	257.4	43.7	101.6	350.0	270.0
μ	5.8	24.3	1018.2	22.1	305.8	56.2	111.5	479.3	311.2
σ	2.0	4.8	123.1	13.3	61.6	13.9	17.8	177.5	54.4

$$\ln \lambda_i = \frac{i-1}{n_g-1} \ln r_a + \ln \lambda_{gmax} \quad (17)$$

where n_g is the total number of grid points (set to 100 in this study); λ_{gmax} is the L1 regularization weight when all coefficients in LASSO regression coefficients nearly zero; r_a is the ratio of the smallest to the largest λ value in the grid. When the number of regression samples is no more than the number of features, $r_a = 0.01$. The initial λ grid were refined by merging non-overlapping parts into a shared grid across all φ values. This approach ensures comprehensive coverage while maintaining computational efficiency.

4.3. Results of ENet regression

Table 5 presents the values of φ and λ corresponding to the minimum loss function in ENet regression, along with the number of nonzero coefficients. The results show that higher φ values, such as $\varphi = 0.18$ for $r_{\theta_h}^R$, enhance feature selection in ENet regression, resulting in models with greater sparsity. For instance, $r_{\theta_h}^R$ has only 5 non-zero regression coefficients. Conversely, lower φ values, like $\varphi = 0.08$ for $gD3$, shift the regression model towards Ridge regression, addressing multicollinearity and yielding more non-zero regression coefficients (7 for $gD3$). The λ values exhibit a wide range from 0.14 to 1.69, indicating considerable variation in regularization strength across different model parameters. This variation in regularization parameters (φ and λ) highlights the inherent complexity of the *Pinching4* model parameters.

Table 6 displays the regression coefficients for standardized features as determined by ENet regression. In this table, feature names are subscripted to indicate that they are standardized, e.g., $n_{t-std} = (n_t - 5.8)/2.0$, where 5.8 and 2.0 are the mean and standard deviation of the original n_t , as shown in the previous Table 4. Standardization is essential for regularization within the ENet loss function, as it ensures all features are on the same scale, preventing potential numerical instability issues. It can be seen that standardized features such as d_{th-std} , $f_{ubo-std}$, and m_{0-std} contribute to most *Pinching4* model parameters. In contrast, the influence of t_{Tf-std} , g_{s-std} , and D_{b-std} is limited to certain model parameters. For example, D_{b-std} is only correlated with $r_{M_h}^R$, $r_{M_{max}}^U$, $\delta K_{lim-unload}$, $gD3$, $gD3$, and $\delta K_{lim-reload}$. Moreover, the importance of standardized features varies significantly across different regression equations, and their influence (positive or negative correlation) is inconsistent. The varied impact of the standardized features on different *Pinching4* model parameters, as evidenced by the diverse regression coefficients, highlights the complex relationships between connection characteristics and *Pinching4* model parameters.

Fig. 10 presents a scatter plot comparing the predicted and observed values for each *Pinching4* model parameter. The horizontal and vertical axes represent the observed and predicted values, respectively, with the diagonal line indicating perfect prediction. Overall, the scatter points cluster around the diagonal, suggesting that the ENet regression predictions are close to the observed values with no significant bias. To evaluate the goodness of fit for the ENet regression equations on the training data, the coefficient of determination (R^2) is used. R^2 is defined as $R^2 = 1 - RSS/TSS$, where RSS is the residual sum of squares of the regression equation on the training data, determined by Eq. (12), and TSS is the total sum of squares, representing the sum of squared differences between each observation and the overall mean. The R^2 values are presented in Table 7. These values range from 0.781 to 0.873, demonstrating a good fit of the ENet regression models for all *Pinching4* parameters. Additionally, the number of non-zero coefficients in the models does not always

Table 5

Regularization parameters and the number of nonzero coefficients.

Pinching4 parameter (xi)	φ	λ	No. of nonzero coef.
$r_{\theta_h}^R (x_1)$	0.18	0.23	5
$r_{M_h}^R (x_2)$	0.12	0.82	6
$r_{M_{max}}^U (x_3)$	0.14	0.57	6
$gK1 (x_4)$	0.12	0.98	6
$gK2 (x_5)$	0.10	0.17	7
$gK3 (x_6)$	0.16	1.36	5
$gK4 (x_7)$	0.12	0.74	6
$\delta K_{lim-unload} (x_8)$	0.12	1.69	6
$gD1 (x_9)$	0.16	0.14	5
$gD2 (x_{10})$	0.18	0.82	5
$gD3 (x_{11})$	0.08	0.18	7
$gD4 (x_{12})$	0.14	0.21	6
$\delta K_{lim-reload} (x_{13})$	0.12	0.93	6

Table 6

Regression coefficients for standardized features determined by ENet regression.

<i>Pinching4</i> parameter(x_i)	Constant term	n_{t-std}	d_{fb-std}	$f_{lubo-std}$	t_{lf-std}	f_{yT-std}	m_{l0-std}	g_{s-std}	D_{b-std}	f_{yb-std}
$r_{\theta_h}^R$ (x_1)	0.426		-0.347	0.044	0.220		0.203			0.060
$r_{M_h}^R$ (x_2)	0.645	-0.221	0.046		0.404	0.164	-0.247		-0.179	
$r_{M_{max}}^U$ (x_3)	0.159	-0.048	0.190	0.042		0.290		-0.158	-0.211	
$gK1$ (x_4)	0.701	0.378	0.257	-0.228		-0.075	-0.411			-0.253
$gK2$ (x_5)	0.876		0.683	-0.901	0.503	-0.625	-0.779	0.332		0.090
$gK3$ (x_6)	0.530		-0.382	-0.136	-0.236	0.108	0.676			
$gK4$ (x_7)	0.674	-0.391	-0.749	0.661			0.357	-0.053		0.889
δK_{lim_unload} (x_8)	0.217	-0.126	0.049		0.235	0.407		0.047	-0.628	0.058
$gD1$ (x_9)	0.737	0.156	0.518	-0.628			-0.616			-0.128
$gD2$ (x_{10})	0.518		-0.819	0.242			-0.130	0.257		0.205
$gD3$ (x_{11})	0.493	0.635		-0.109	0.262	-0.341	0.210	-0.751	-0.170	
$gD4$ (x_{12})	0.851	-0.208		0.299			0.569	-0.964	0.544	-0.168
δK_{lim_reload} (x_{13})	0.199	0.399	-0.048		-0.720	-0.751	0.217		0.851	

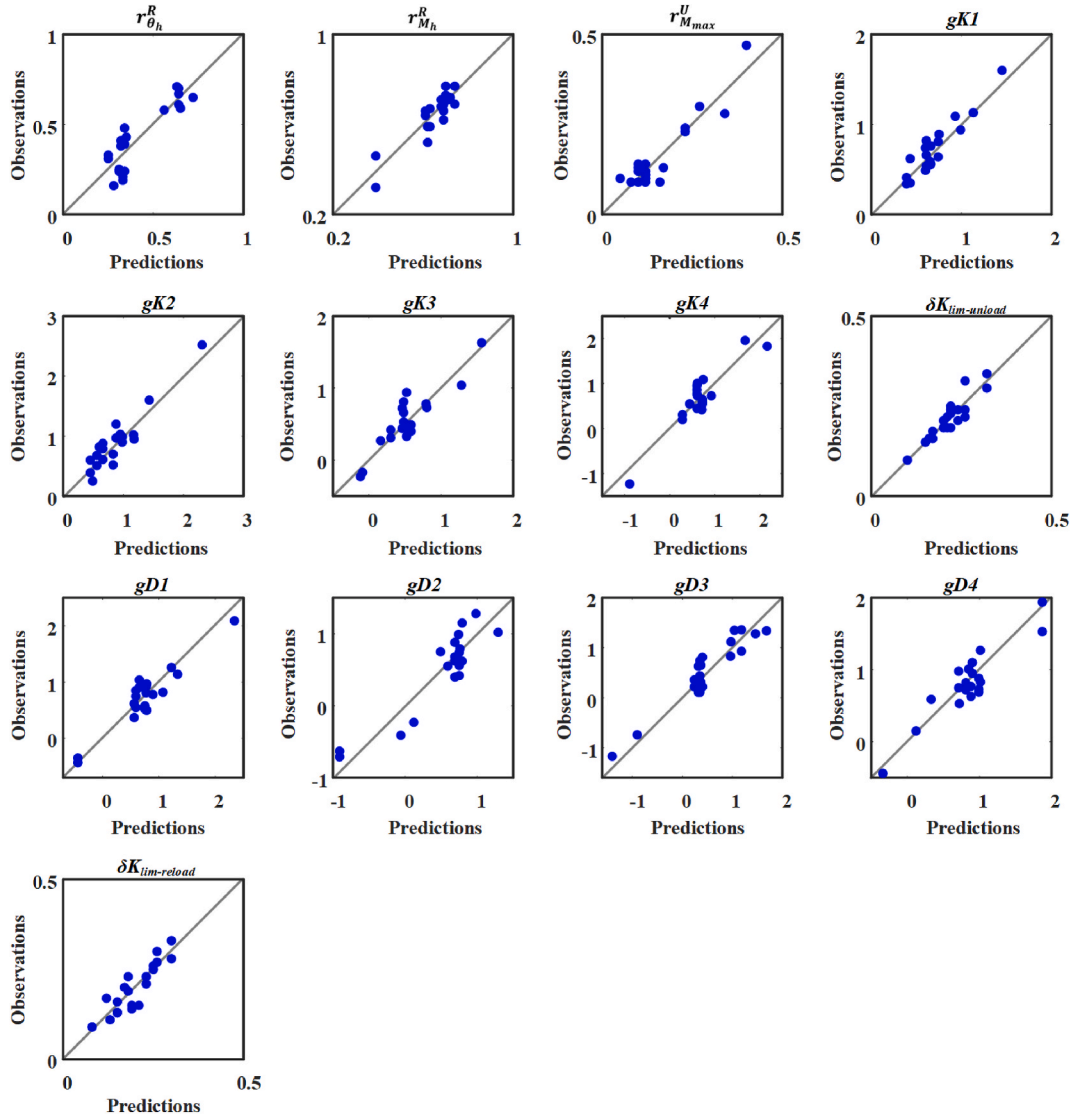
**Fig. 10.** Comparison between the predicted and observed values for each *Pinching4* model parameter.

Table 7Comparison of R^2 with R_{LOOCV}^2 for ENet regression equations.

<i>Pinching4</i> parameter (x_i)	R^2	R_{LOOCV}^2	$ R_{LOOCV}^2 - R^2 / R^2 (\%)$
$r_{\theta_h}^R (x_1)$	0.781	0.613	21.5
$r_{M_h}^R (x_2)$	0.816	0.689	15.6
$r_{M_{max}}^U (x_3)$	0.860	0.627	27.1
$gK1 (x_4)$	0.858	0.698	18.6
$gK2 (x_5)$	0.867	0.721	16.8
$gK3 (x_6)$	0.812	0.696	14.2
$gK4 (x_7)$	0.851	0.715	16.0
$\delta K_{lim, unload} (x_8)$	0.834	0.706	15.4
$gD1 (x_9)$	0.850	0.673	20.8
$gD2 (x_{10})$	0.827	0.650	21.4
$gD3 (x_{11})$	0.873	0.722	17.3
$gD4 (x_{12})$	0.833	0.664	20.3
$\delta K_{lim, reload} (x_{13})$	0.806	0.672	16.6

positively correlate with R^2 . For instance, $gD1$ has 5 non-zero coefficients and an R^2 of 0.850, while $\delta K_{lim, reload}$ has 6 non-zero coefficients and an R^2 of 0.806. This indicates that ENet regression effectively balanced model complexity (in terms of the number of non-zero coefficients) with predictive performance.

Considering the small sample size (20 specimens) used in ENet regression, there is a potential risk of overfitting. While R^2 reflects the performance of the regression equations on the full training dataset, it is unable to assess generalizability. To address this, the predictive coefficient of determination based on leave-one-out cross-validation, denoted as R_{LOOCV}^2 , is further calculated. It is defined as $R_{LOOCV}^2 = 1 - PRSS / TSS$, where $PRSS$ represents the predictive residual sum of squares from LOOCV. $PRSS$ is calculated using Eq. (12), with $x_{i,pre}^{(j)}$ redefined as the prediction for the j -th sample by training the ENet regression model on the remaining samples (excluding the j -th sample). The computation of TSS remains the same as in R^2 . The R_{LOOCV}^2 values are also presented in Table 7. It is observed that, for all ENet regression equations, R_{LOOCV}^2 are lower than their corresponding R^2 values by approximately 14.2 %–27.1 %. This indicates that although the predictive performance of the ENet regression equations decreases to some extent in cross-validation, no significant overfitting is observed. The regression equations still maintain reasonable generalization capability. Nevertheless, expanding the experimental database in future would further enhance the generalizability and accuracy of the ENet regression equations.

It is worth noting that while the prior distributions for the *Pinching4* model parameters defined in this study theoretically cover a broad range of steel BCBT connections with diverse geometric dimensions and material properties, the experimental dataset used for calibration and subsequent ENet regression is relatively limited. If new steel BCBT connections significantly exceed the scope of current database (e.g., with distinct bolt sizes or material strengths), the predictions of the developed regression equations may involve extrapolation risks. In such cases, experimental validation may be necessary. Nonetheless, this study provides an initial framework for predicting hysteretic model parameters for steel BCBT connections without relying on extensive test data. It is not intended to replace large-scale experimental studies but rather to complement them by offering a systematic methodology for parameter estimation.

5. Conclusions

This study presents a “physics-informed + data-driven” framework to predict the nonlinear moment-rotation behavior of steel BCBT connections under limited data conditions. The *Pinching4* model in *OpenSees* is employed to simulate the hysteretic curves under cyclic loading. An approach integrating PCE and GA is proposed to identify the *Pinching4* model parameters, and ENet regression then establishes predictive equations for these parameters. This framework offers distinct advantages in terms of handling small dataset, physical interpretability, and engineering applicability. The main findings are as follows.

- (1) **Surrogate model-based calibration:** The surrogate model-based approach demonstrates significant potential in calibrating the *Pinching4* model parameters for steel BCBT connections. The PCE surrogate models show stable predictive performance across different specimens, achieving good accuracy with leave-one-out cross-validation errors ranging from 0.035 % to 0.064 %. The optimal truncated order, which balances the complexity and predictive accuracy of the PCE models, is found to be either 3 or 4.
- (2) **GA optimization results:** GA optimization successfully identifies parameter sets that minimize the discrepancy between the *Pinching4* model predictions and experimental results. For 15 out of 20 specimens, final $RRMSE$ lies between 0.091 and 0.195, indicating close agreement with experimental results. Slightly larger errors (0.233–0.289) occur for five specimens with significant hysteresis asymmetry or experimental noise.
- (3) **ENet regression for parameter prediction:** ENet regression effectively addresses the challenges of high-dimensional features and small sample sizes. The regression equations for the 13 *Pinching4* model parameters demonstrate favorable predictive performance, with R^2 values ranging from 0.781 to 0.873, showing suitable accuracy while maintaining interpretability. A modest drop (14.2 %–27.1 %) in leave-one-out cross-validation R^2 values indicates limited overfitting.

The regression equations developed for the 13 *Pinching4* model parameters, together with the predictive models for the cyclic envelopes in an earlier work, enable practical use of the *Pinching4* model in modeling the moment-rotation hysteretic behavior of steel BCBT connections when experimental data is unavailable.

CRedit authorship contribution statement

Yanfei Shen: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Mao Li:** Writing – review & editing. **Yong Li:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix

Table A1

A summary of all possible parameters in the *Pinching4* model

Direction	Envelope parameters (force, deformation)	Pinching shape	Unloading stiff. degradation	Reloading stiff. (in- cycle strength) degradation	Cyclic strength degradation	Max. energy dissipation parameter	Damage type
Positive	(<i>ePf1</i> , <i>ePd1</i>), (<i>ePf2</i> , <i>ePd2</i>), (<i>ePf3</i> , <i>ePd3</i>), (<i>ePf4</i> , <i>ePd4</i>)	$rDispP(r_{\theta_b}^R)$, $rForceP(r_{M_b}^R)$, $uForceP(r_{M_{max}}^U)$	$gK1$, $gK2$, $gK3$, $gK4$, $gKLim$ ($\delta K_{lim, reload}$)	$gD1$, $gD2$, $gD3$, $gD4$, $gDLim$ ($\delta K_{lim, unload}$)	$gF1$, $gF2$, $gF3$, $gF4$, $gFLim$	gE	<i>dmg</i> <i>Type</i>
Negative	(<i>eNf1</i> , <i>eNd1</i>), (<i>eNf2</i> , <i>eNd2</i>), (<i>eNf3</i> , <i>eNd3</i>), (<i>eNf4</i> , <i>eNd4</i>)	$rDispN(r_{\theta_b}^R)$, $rForceN(r_{M_b}^R)$, $uForceN(r_{M_{max}}^U)$					

Note: Symmetry is assumed for both positive and negative loading; the original parameter notations in the model — $rDispP(rDispN)$, $rForceP(rForceN)$, $uForceP$ ($uForceN$), $gKLim$ and $gDLim$ — correspond to $r_{\theta_b}^R$, $r_{M_b}^R$, $r_{M_{max}}^U$, $\delta K_{lim, reload}$ and $\delta K_{lim, unload}$, respectively, in this study.

Table A2

Characteristic points of the cyclic envelopes for 20 steel beam-column bolted T-stub connections

Specimen	M_y (MPa)	θ_y (rad)	M_{max} (MPa)	θ_c (rad)	M_r (MPa)	θ_r (rad)
B4BC1-E	107.5	0.0017	284.4	0.0158	NA	NA
B4BC3-E	209.0	0.0021	414.4	0.0176	NA	NA
SS-02a	801.7	0.0021	1221.0	0.0293	1149.0	0.0328
SS-02b	785.8	0.0017	1219.4	0.0225	1080.0	0.0251
FS-06	879.0	0.0029	1008.9	0.0258	936.0	0.0361
FS-07	790.9	0.0027	1077.5	0.0243	820.8	0.0379
FS-08	796.9	0.0026	963.6	0.0207	760.1	0.0219
PJD1-N	84.0	0.0125	133.0	0.1040	NA	NA
PJD1-S	88.7	0.0135	120.6	0.0910	110.5	0.1284
PJD2-N	88.3	0.0095	153.4	0.1110	NA	NA
PJD2-S	95.5	0.0100	156.0	0.1240	NA	NA
Specimen 1	4357.0	0.0018	5188.4	0.0100	NA	NA
Specimen 2	4221.7	0.0016	4932.9	0.0067	3493.0	0.0074
KJD1-N	59.4	0.0115	83.1	0.0770	78.1	0.0900
KJD1-S	62.2	0.0075	88.5	0.0690	45.0	0.1239
KJD2-N	67.4	0.0120	110.6	0.0930	NA	NA
JATD12-N	66.0	0.0110	96.6	0.1217	78.7	0.1400
JATD12-S	71.3	0.0076	106.4	0.0845	67.6	0.0961
JATD14-N	92.1	0.0120	127.4	0.0767	94.0	0.1033
JATD14-S	61.6	0.0055	98.6	0.0435	NA	NA

Note: M_y , M_{max} and M_r represent the yielding moment, maximum moment, and residual moment, respectively, while θ_y , θ_c and θ_r correspond to the rotations at M_y , M_{max} and M_r , respectively; NA = not applicable; *Pinching4* envelope points (*ePf1*, *ePd1*), (*ePf3*, *ePd3*), and (*ePf4*, *ePd4*) correspond to (M_y , θ_y), (M_{max} , θ_c), and (M_r , θ_r), respectively; and the values of *ePf2* and *ePd2* are determined by linear interpolation.

Table A3
Prior Gaussian distributions for the *Pinching4* model parameters

<i>Pinchin4</i> parameter (x_i)	99.74 % Confidence interval		μ_{prio}	σ_{prio}
	b_u	b_l		
$r_{fb}^R(x_1)$	0.99	0.01	0.50	0.16
$r_{Mb}^R(x_2)$	0.99	0.01	0.50	0.16
$r_{Mmax}^U(x_3)$	0.99	0.01	0.50	0.16
$gK1(x_4)$	5.00	−5.00	0.00	1.67
$gK2(x_5)$				
$gK3(x_6)$				
$gK4(x_7)$				
$\delta K_{lim,unload}(x_8)$	0.99	0.01	0.50	0.16
$gD1(x_9)$	5.00	−5.00	0.00	1.67
$gD2(x_{10})$				
$gD3(x_{11})$				
$gD4(x_{12})$				
$\delta K_{lim,reload}(x_{13})$	0.99	0.01	0.50	0.16

Note: b_u and b_l represent the upper boundary and the lower boundary, respectively; μ_{prio} and σ_{prio} are mean and standard deviation, respectively.

Data availability

Data will be made available on request.

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